



AMERICAN INTERNATIONAL UNIVERSITY - BANGLADESH

Project Proposal
**E-Commerce Sales Forecasting Using Machine Learning and
Interactive Visualization**

Course: Research Methodology

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Submitted by

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1. Abstract

A great deal of data is collected by e-commerce firms on sales, items, clients, costs, discounts, regions, stocks and return rates. In turn, analysis of such data allows to identify sales patterns and predict future sales, making it possible to better plan inventory, prices, promotions and product development. This study suggests the implementation of the E-Commerce Sales Forecasting System using Machine Learning and Interactive Visualization. Firstly, it is planned to gather and cleanse historical transaction data from e-commerce. Secondly, it is planned to conduct feature engineering and Exploratory Data Analysis (EDA) in order to identify essential patterns, trends and interrelations between variables regarding the sales. At the same time, two different machine learning algorithms will be constructed and compared in order to forecast future sales – Random Forest and XGBoost. Models' accuracy will be measured by Mean Absolute Error (MAE), Root Mean Squared Error (RMSE) and R^2 . A time-series method will be used for splitting the data set into training and test sets to make the forecasting more appropriate for historical sales data. Finally, forecast results will be integrated into the interactive visualization. It will include key sales indicators, sales trends, performance of products and regions, analysis of clients, profits and discounts, comparison of models and sales forecasts for the future periods.

2. Proposal Keywords

E-Commerce, Sales Forecasting, Machine Learning, Random Forest, XGBoost, Predictive Analytics, Interactive Visualization

3. Introduction and Problem Statement

The development of e-commerce has led to the emergence of large amounts of data related to transactions and customers. The data can include orders, goods, prices, discounts, clients, region, inventories, refunds, and amounts of sales. Correct analysis of such data will allow identifying customer behavior, tendencies of sales, goods' performance, and future demand.

Sales forecasting is necessary since it can be used for inventory planning, demand forecasting, pricing, promotional planning, resource allocation, and business strategy in general. The classical methods of statistical forecasting like time-series forecasting can be helpful in understanding time trends and dependencies. Modern machine learning algorithms are able to analyze relations between several variables influencing sales. A systematic review of the literature related to sales forecasting revealed that there are many forecasting approaches and their choice depends on forecasting specifics and data [1].

Different researchers analyzed machine learning and deep learning models for sales and demand forecasting. These were Random Forest, XGBoost, LightGBM, CatBoost, LSTM, GRU, and other approaches [2]-[10].

Sales through e-commerce channels are likely to be inconsistent and could differ based on seasonality, prices, discounts, customer preferences, characteristics of the products, and regional specifics. Another issue is related to the dependence of forecasting performance on a number of factors such as dataset, selected features, the forecast period, and algorithms used. It was shown that various forecasting results can be obtained using machine learning, statistical, and deep learning models [1], [7]–[10].

At the same time, numerous studies concentrate on forecasting, although business people also require comprehensible information on products, regions, customers, profits, discounts, and inventory. Likewise, interactive dashboards facilitate the understanding of business data but could have little or no forecasting at all [13], [14].

3.1. Research Gap

The base Paper for Research: "*Grocast: Interactive Sales Forecasting Dashboard*" [14]. This work introduces an interactive Power BI sales dashboard meant for KPI measurement, trends analysis, sales performance analysis, and decision-making based on data.

The authors have pointed out advanced analytics, artificial intelligence, machine learning, automated insights, anomaly detection, and predictive analytics as possible future work.

1. Machine Learning Model Comparison: No comparison of different ML models for forecasting sales is provided by Grocast.
2. Limitations in Explanation of Factors: The dash does not explain the factors that affect sales forecasts.
3. Limitations in Generation of Business Insights: Forecasting is not entirely integrated with analysis of products, customers, regions, and profits.
4. Limitations in Decision-Making: Information, rather than business insights for decision-making is offered.
5. Necessity of Integration of a Framework: It is required to integrate ML forecast and its comparison, explanation of factors, and visualization into one system.

4. Literature Review

The field of sales forecasting research has evolved from statistical methods to machine learning, ensemble learning, and deep learning methods. Ahaggach et al. have conducted a systematic review of the sales forecasting methods and emphasized the variety of the forecasting methods and the selection of suitable methods based on the forecasting challenge [1].

Traditional methods of forecasting still prove to be helpful for time-dependent sales. Machine learning models are able to find nonlinear interrelations between several factors. Feng et al. have conducted research of the short-term sales forecasting in e-commerce by means of machine learning models and proved the usefulness of machine learning in sales forecasting in e-commerce [2]. Lei has analyzed sales forecasting in cross-border e-commerce platforms by means of machine learning [3]. Singh et al. have developed an e-commerce system using machine learning algorithms for predicting sales [4]. The more recent work by Singh discusses more sophisticated machine learning algorithms for better forecasting of sales in e-commerce

businesses [5]. Venkiteela has also introduced a machine learning framework for retail sales forecasting using machine learning [6].

Comparative analysis is also significant since there may be no universal model for every case of forecasting. Sun compares traditional models and machine learning methods in terms of e-commerce sales forecasting and their performance [7]. Subashree et al. compare LightGBM, CatBoost, and Prophet algorithms in sales and demand forecasting [8].

Deep learning has emerged as another important research area. Mekala et al. studied e-commerce sales forecasting through deep learning [9]. Ecevit et al. conducted a study on short-term sales forecasting with the help of LSTM, Prophet, and similar models [10]. Chen et al. developed a model based on GRU-LightGBM for e-commerce sales forecasting in short-shelf-life items [16]. Moreover, Moraes et al. also worked on the hybrid approach of convolution and LSTM models for retail sales forecasting [17].

Moreover, sales forecasting has some relation with inventory management and supply chain management. Yusof worked on the topic of predictive analytics and machine learning in e-commerce inventory management and demand forecasting [11]. In another paper, Bucha studied machine learning and deep learning methods for e-commerce supply chain forecasting and the challenges of computational cost, data dependency, scalability, and interpretation [12]. Another area worth researching is that of the integration of forecasting with business intelligence and visualization. Al-Etaibi et al. studied data-based prediction and insights into e-commerce with the use of machine learning and business intelligence solutions [13]. The examples prove the opportunities for integrating predictive analytics and business intelligence to help in decision making.

Januschowski et al. analyzed classification of forecasting methods and noted that forecasting methods must be chosen depending on features of forecasting problems [15]. Fu gave a review of deep learning for sales forecasting and discussed deep learning forecasting development and challenges [19]. Panda et al. discussed application of AI and machine learning in e-commerce and mentioned sales forecasting and product recommendation application as well [18].

5. Research Objectives

1. To collect data, clean and analysis of historical e-commerce transaction data to find sales trends and patterns.
2. To identify relevant sales-related features such as Quantity, Price, Discount, Profit, Product, Region, Customer and Time- Related Features.
3. To develop Random Forest and XGBoost models to Forecast Future E-commerce Sales.
4. To compare the forecasting performance of Random Forest and XGBoost using Metrics such as MAE, RMSE and R^2 .
5. To develop an interactive dashboard to presenting KPIs of Sales, Trends, Performance by Products and Regions, Profit Analysis and Future Sales Forecasts.
6. To provide Business Decision Support Insights on Demand Planning, Inventory, Pricing, Promotions and Products Strategy.

5.1. Possible Novelty

1. ML Model Comparisons: Random Forest vs XGBoost for sales forecasting.
2. Explainable Forecasting: Highlight major factors influencing sales forecasts.
3. Integrated Dashboard: Integration of forecasting and analysis of product, customer, regional, and profitability metrics.
4. Business Insights: Generate useful information on inventory, pricing, marketing, and product development.
5. Comprehensive Approach: Integration of ML forecasting, explainability, analysis, and visualization.

6. Proposed Methodology

The proposed research will follow the following workflow:

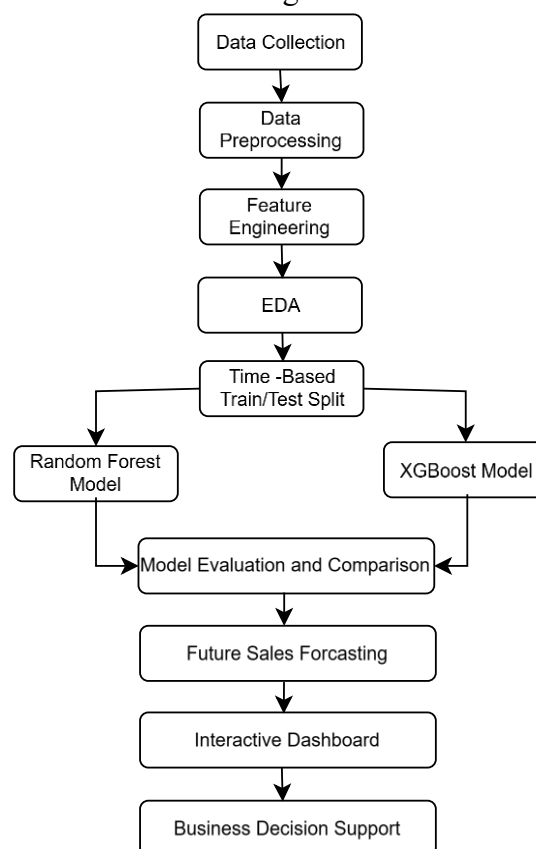


Fig: The proposed project workflow

6.1. Dataset

The suggested research project will make use of the past e-commerce transactional data in order to study the sales trend and predict future sales. Such variables as Order ID, Selling Price, Customer ID, Cost Price, Order Date, Discount, Product, Sales Amount, Region, Stock, Quantity, and Returned Orders will make the key part of the data that is to be studied. These variables are going to be studied in order to reveal sales trends, product and region performance, customer purchasing pattern, impact of pricing and discounts on the sales, profits, and stock trends.

6.2. Data Preprocessing

Before applying machine learning algorithms, data from the online store will be verified and preprocessed to make sure that the dataset can be analyzed. The following steps of data validation and preprocessing will be undertaken:

- **Missing Data:** Missing or blank data will be found and processed accordingly.
- **Duplicate Data:** Duplicate transactions will be detected and deleted to avoid duplication of data.
- **Incorrect Data Type:** Cases when data are in wrong type, for example, string value in numeric field, will be found and fixed.
- **Incorrect Date:** Incorrect date values will be found and corrected or removed if needed.
- **Outliers:** Outlier values in sales amount, price, quantity and profit variables will be found and analyzed.

6.3. Feature Engineering

Once the data is preprocessed, relevant features are generated out of the given e-commerce transaction data. Time-based features like year, month, and season are generated in order to find any changes in sales that happen over time. A rolling average of sales is generated in order to know about the latest trend of sales. Profit, profit status and profit margins are computed in order to analyze the profitability of sales. Average Order Value (AOV) is computed in order to find out how much on an average is spent per order. Discount-based features are generated in order to analyze the impact of discounts on sales. In addition to this, features are generated based on products and regions to analyze the differences in sales performance across products and regions.

6.4. Exploratory Data Analysis

The Exploratory Data Analysis (EDA) will be conducted to determine the key patterns and trends that occur within the e-commerce dataset before developing machine learning models. This will include analyzing the overall trend of sales and the seasonality of sales in the business. The product performance will be evaluated to determine the products with high and low sales. At the same time, regional performance will be assessed to understand regional variations in sales. Furthermore, customer behavior will be evaluated to determine any patterns. Moreover, the association between discount and sales performance will be identified through the EDA process. Finally, profit and the profit margin will be assessed as part of EDA.

6.5. Machine Learning Models

Two models will be developed:

- **Random Forest** will be utilized for predicting the future aggregated sales based on chosen features from historical data, temporal variables, product characteristics, region, price, and promotions.
- **XGBoost** is a boosting algorithm which is developed for accurate and efficient predictions on structured data. It trains the model by building a chain of decision trees.

6.6. Model Evaluation

The models will be evaluated based on:

Metric	Desired Outcome
MAE	Lower
RMSE	Lower
R ²	Higher

- Mean Absolute Error (MAE) represents the average absolute difference between actual and predicted values.

$$MAE = (1/n) \sum_{i=1}^n |y_i - \hat{y}_i|$$

- Root Mean Square Error (RMSE) measures the average size of prediction errors while giving more weight to larger errors.

$$RMSE = \sqrt{[(1/n) \sum_{i=1}^n (y_i - \hat{y}_i)^2]}$$

- R-Squared (R²) represents how much of the variation in the target variable is explained by the model.

$$R^2 = 1 - [\sum_{i=1}^n (y_i - \hat{y}_i)^2 / \sum_{i=1}^n (y_i - \bar{y})^2]$$

6.7. Interactive Dashboard

The visualization will take the form of an interactive dashboard that will enable easy presentation of the results from sales analysis and forecasting. Key aspects that will be covered by the dashboard include sales trends and future sales forecasts. Product and regional performance analysis will also be available so that it is possible to determine the relative success of different products and regions. Another key area of focus will be customer analysis so that customer purchase trends can be determined. Apart from that, the profit and profit margins will also be presented in the dashboard. Discount analysis will also be available so that the relationship between discounts and sales can be determined.

6.8. Business Decision Support

- Demand Planning: To know about the future demand for sales and be prepared for the anticipated customer demand.
- Inventory Management: To know about the stock requirement considering the sales trend and forecasted sales.
- Pricing: To make effective pricing decisions considering the impact of price, discount, and sales.
- Promotion: To analyze the impact of discounts and promotions on sales performance.
- Product Strategy: To find out high and low performing products for making product related decisions.
- Regional Strategy: To analyze the sales performance in different regions and make regional business decisions.

7. Expected Outcomes and Contributions

The proposed research will help:

1. Discover valuable historical patterns of e-commerce sales.
2. Forecast future sales by means of machine learning.
3. Compare Random Forest and XGBoost forecast performance.
4. Select the superior forecasting method.
5. Analyze product, region, price, discount, and profitability patterns.
6. Design an interactive sales forecasting dashboard.
7. Present sales forecasts in an easily understood manner.
8. Support decision-making regarding inventory, pricing, promotions, and products.

8. Project Timeline

Project Activities	W 1	W 2	W 3	W 4	W 5	W 6	W 7	W 8	W 9	W 10	W 11	W 12
Finalized Topic, Problem Statement and Objectives												
Literature Review and Research Gap												
Dataset Collection and Understanding												
Data Preprocessing												
Feature Engineering and EDA												
Random Forest Model Development												
XGBoost Model Development												
Model Evaluation and Comparison												
Dashboard Design												
Forecasting and Dashboard Integration												
Business Analysis and Testing												
Final Report												

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